

NYASHA PRAISE MAPETERE

JUNIOR SOFTWARE ENGINEER | TECHNICAL ACCOUNT SPECIALIST | CYBERSECURITY ANALYST

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PROFESSIONAL SUMMARY

Computer Engineer with a First Class Degree and experience in secure software development, DevOps and systems architecture. Deeply focused on cybersecurity, threat mitigation and building resilient infrastructure. Known for being analytical, security-conscious and quick to adapt, I excel at solving complex vulnerabilities and engineering solutions that prioritize data integrity and system reliability.

PROFESSIONAL EXPERIENCE

Technical Account Specialist | Iteb Co

Jul 2026 – Present

- Conducted comprehensive technical and user experience audits for clients such as Rise On Tours, leveraging Google Lighthouse for performance profiling, mobile responsiveness testing and SEO checking.
- Delivered detailed, prioritized assessment reports bridging technical web development findings with strategic business growth recommendations.
- Evaluated design, content and UI elements to optimize customer journeys, improving overall conversion potential and user engagement for tourism clients.

Software Engineering Intern (QA & Systems Analyst) | Afrosoft

Sep 2024 – Aug 2025

- Architected enterprise grade pipelines in Jenkins for auto deployment in Java projects like the NRZ Biometrics System, improving system efficiency by 45%.
- Functioned as the sole QA Engineer for the ZDF Loans and Finance Manager system (React, Java, PostgreSQL), executing rigorous integration testing via JUnit and Docker to guarantee a flawless, on time production deployment.
- Translated complex, non technical user requirements into structured technical specifications, transforming client needs into testable Java centric backend business logic.
- Spearheaded end to end task tracking, project management and cross functional milestone coordination for the ZDF project modules to meet strict institutional deadlines.
- Collaborated with senior architects to ensure 100% compliance with financial security standards for the ZDF Loans and Finance Manager System files.

SELECTED TECHNICAL PROJECTS

ZET 5 Automated Energy Priority System (Final Year Dissertation — Awarded Distinction)

- Designed and developed a Software in the Loop (SIL) energy tracking and predictive load shedding controller utilizing modern JavaScript, React 19.1, Vite, Node.js, WebSockets and LocalStorage.
- Engineered an offline, edge hosted React dashboard that updates every 1.5 seconds independently of cloud reliance.
- Implemented an Exponential Moving Average (EMA) algorithm to create a Hybrid Reconciliation Loop, reducing simulated hardware integration drift from 15% to 0.4%.
- Developed an O(1) time complexity priority engine that autonomously preserves essential loads during token depletion.

Intelligent OBD II Edge Diagnostic System (Engineering Design & Make Project)

- Architected an offline first vehicle edge node in Rust utilizing Tokio and SocketCAN, achieving a strict 10Hz real time CAN bus polling deadline and a microscopic 1.84MB memory footprint while processing over 489,000 frames without data loss.
- Engineered a cross platform Flutter/Dart mobile application featuring live 3D vehicle rendering and WebSocket telemetry streaming, leveraging local SQLite for massive offline drive logging.
- Designed embedded hardware integration linking a Raspberry Pi 5 to an MCP2515 CAN HAT via SPI; managed PCB layout and fabrication processes with KiCAD and simulated electrical thresholds using LTSpice.
- Configured the Raspberry Pi environment for local control via a direct connected monitor interface and administered network configurations securely over SSH.
- Secured AWS cloud based Python batch processing for unsupervised ML feature discovery by implementing Ed25519 elliptic curve digital signatures, comprehensively preventing malicious code injection.

LANGUAGES & FRAMEWORKS

Python (Flask/Django) Rust Java (Spring Boot)
JavaScript (React 19.1/Vite) TypeScript Dart (Flutter) PHP
SQL (PostgreSQL/MySQL/SQLite) Tailwind CSS Supabase Tokio
WebSockets

CORE TOOLS & INFRASTRUCTURE

WBS Deployment Cloud Object Storage Ed25519 Cryptography
LocalStorage Raspberry Pi KiCAD LTSpice SSH Docker
Terraform Git/GitHub Jenkins MongoDB Cloudflare
Linux/UNIX Admin Linux SocketCAN OBD II/CAN Bus SPI

CERTIFICATIONS

Introduction to Cybersecurity (Cisco - 2026)
DevOps: Beginner to Advanced (Udemy)
Introduction to Java (Oracle / LinkedIn)

EDUCATION & TRAINING

BEng (Hons) Computer Engineering (1st Class)

Chinhoyi University of Technology
Graduating August 2026

Intensive Software Training

She Code

March 2026

Focused on Frontend Designing.

A Level | Thornhill High, Gweru

3 A Level Passes: Computer Science, Mathematics, Physics

O Level | Regina Mundi High, Gweru

12 O Level Passes incl. Computer Science, Mathematics, English

REFERENCES

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